Math Inquirers Level 4: skill additions (tutor sheet)

Extra items for early finishers and for tutors to set live. They are additions: nothing here is in the printed workbook. **Slide** items are on an optional slide in the lesson deck (just before Talk time). **Bank** items are the same skill with new numbers: read or write them for the pair. Children write the answer on their own paper and say it together on three. Nothing is held up to the camera. Never announce who is working on which item.

Lesson 1: Factor Hunters

SLIDE Tool: Spot the pattern **Digit tests for 3 and 6** (L4-DIGIT)

(a) Add the digits of each number. Which of 724 and 618 is a multiple of 3? A number in the 2 table and in the 3 table is also in the 6 table. (b) The 3-digit number 58[ ] is a multiple of 6. Which digits can go in the box? Write every answer.

Answer. (a) 724: digit sum 13, no. 618: digit sum 15, yes. (b) The box can be 2 or 8: 582 and 588.

Say. Look at a multiple of 3 such as 51, 63, 84. Add their digits. What do you notice? Then try 724 and 618.

Ask. If a number is in both the 2 table and the 3 table, what must be true about its last digit and its digit sum?

Listen for. The digit sum of a multiple of 3 is a multiple of 3. Even last digit AND digit sum in the 3 table. Children who test all 10 digits are fine, but ask them to find a faster way.

BANK Tool: Spot the pattern **Digit tests for 3 and 6** (L4-DIGIT-V)

The 3-digit number 73[ ] is a multiple of 6. Which digits can go in the box?

Answer. 732 and 738 (box 2 or 8).

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Digit tests for 3 and 6).

Ask. If a number is in both the 2 table and the 3 table, what must be true about its last digit and its digit sum?

Listen for. The digit sum of a multiple of 3 is a multiple of 3. Even last digit AND digit sum in the 3 table. Children who test all 10 digits are fine, but ask them to find a faster way.

SLIDE Tool: Make a table **Weekdays and remainders** (L4-WEEKDAY)

(a) Ben borrows a library book on a Monday and must return it 45 days later. Name the weekday of the return date. (b) Kumar starts a trip on a Saturday. Name the weekday that comes 30 days after that Saturday.

Answer. (a) Thursday. (b) Monday.

Say. A week has 7 days, so the days repeat in a cycle of 7. Write the days across the top of a table, then jump 7 days at a time.

Ask. How many whole weeks are in 45 days? What is left over, and what does the leftover tell us?

Listen for. 45 = 6 × 7 + 3, so it is the same as 3 days later. This is the remainder move from today's lesson. Listen for "divide by 7 and look at the remainder".

BANK Tool: Make a table **Weekdays and remainders** (L4-WEEKDAY-V)

Siti starts swimming lessons on a Tuesday. Name the weekday that comes 100 days after that Tuesday.

Answer. Thursday (100 = 14 weeks and 2 days).

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Weekdays and remainders).

Ask. How many whole weeks are in 45 days? What is left over, and what does the leftover tell us?

Listen for. 45 = 6 × 7 + 3, so it is the same as 3 days later. This is the remainder move from today's lesson. Listen for "divide by 7 and look at the remainder".

Lesson 2: Big Number Puzzles

SLIDE Tool: Guess and check **Letter sums** (L4-CR1)

Each letter stands for one digit. Different letters stand for different digits. A letter at the start of a number is not 0. Find P, Q, R and S.

Answer. P = 5, Q = 1, R = 9, S = 0. Check: 51 + 59 = 110.

Say. Two 2-digit numbers cannot add to more than 198. So what can the first digit of the answer be?

Ask. Which letter is that digit? What does the tens column then tell you about P?

Listen for. Q = 1 from the size of the sum, then 2 x P + carry gives a number ending in 1 with a carry, so P = 5. Children who guess P first and test it are using Guess and check well.

BANK Tool: Guess and check **Letter sums** (L4-CR1-V)

Each letter is a different digit. TT + UT = TVW. Find T, U, V and W.

Answer. T = 1, U = 9, V = 0, W = 2 (11 + 91 = 102).

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Letter sums).

Ask. Which letter is that digit? What does the tens column then tell you about P?

Listen for. Q = 1 from the size of the sum, then 2 x P + carry gives a number ending in 1 with a carry, so P = 5. Children who guess P first and test it are using Guess and check well.

SLIDE Tool: Guess and check **Letter sums, a longer one** (L4-CR2)

Each letter stands for one digit. Different letters stand for different digits. A letter at the start of a number is not 0. Find D, A, E, C and B.

Answer. D = 8, A = 3, E = 1, C = 4, B = 0. Check: 883 + 148 = 1031.

Say. Two 3-digit numbers add to a 4-digit number. What is the biggest the first digit of the answer can be? Start there.

Ask. Which letter is the answer's first digit, and what must the hundreds column be to make it?

Listen for. E = 1 first. Then children list possibilities for the hundreds column (D + E + carry) and test with the other columns. A table with one column per letter helps.

Lesson 3: Operation Detectives

SLIDE Tool: Try something simpler **Adding a list by pairing** (L4-PAIR)

Here is a list: 11, 14, 17, 20, ... , 41. The list goes up in steps of 3. (a) How many numbers are in the list? (b) Find the sum of all the numbers.

Answer. (a) 11 numbers. (b) 286.

Say. Try something simpler: write the first and last number. Add them. Then the second and the second last. What do you see?

Ask. What is the same about every pair? How many pairs are there, and what is left over in the middle?

Listen for. Every pair adds to 52 (11 + 41, 14 + 38, ...). 5 pairs and a middle number 26. Listen for "first plus last, second plus second last".

BANK Tool: Try something simpler **Adding a list by pairing** (L4-PAIR-V)

Find the sum of 5, 9, 13, ... , 49.

Answer. 324 (12 numbers, 6 pairs of 54).

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Adding a list by pairing).

Ask. What is the same about every pair? How many pairs are there, and what is left over in the middle?

Listen for. Every pair adds to 52 (11 + 41, 14 + 38, ...). 5 pairs and a middle number 26. Listen for "first plus last, second plus second last".

SLIDE Tool: Make a table **Gaps that grow** (L4-GAP)

Ravi saves coins in a jar. Day 1: 3 coins in the jar. Day 2: 5. Day 3: 8. Day 4: 12. Day 5: 17. The number he adds goes up by 1 each day. (a) How many coins are in the jar on Day 6? (b) How many coins are in the jar on Day 10?

Answer. (a) 23 coins. (b) 57 coins.

Say. Make a table with three columns: the day, the number of coins, and the gap from the day before.

Ask. What is the gap on Day 2? On Day 3? What will the gap be on Day 6?

Listen for. Gaps 2, 3, 4, 5, then 6, 7, 8, 9, 10. Listen for "the gap goes up by 1" and a table that keeps going without skipping a row.

BANK Tool: Make a table **Gaps that grow** (L4-GAP-V)

Mei folds paper stars. Day 1: 4, Day 2: 6, Day 3: 9, Day 4: 13. The number added goes up by 1 each day. How many on Day 8?

Answer. 39 (the days go 4, 6, 9, 13, 18, 24, 31, 39).

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Gaps that grow).

Ask. What is the gap on Day 2? On Day 3? What will the gap be on Day 6?

Listen for. Gaps 2, 3, 4, 5, then 6, 7, 8, 9, 10. Listen for "the gap goes up by 1" and a table that keeps going without skipping a row.

Lesson 6: Angle Turns

SLIDE Tool: List every way **Triangles by size** (L4-FAN)

The top point is joined to 4 points on the base line. (a) Count the triangles with 1 small triangle inside, with 2, and with 3. (b) How many triangles are there altogether?

Answer. (a) 3 triangles of size 1, 2 of size 2, 1 of size 3. (b) 6 triangles altogether.

Say. Name the base points B, C, D, E. A triangle needs the top point and two base points. List the pairs in order.

Ask. How do you know you have not missed one or counted one twice?

Listen for. An ordered list BC, BD, BE, CD, CE, DE, or a count by size 1, 2, 3. Listen for a rule such as "choose 2 base points".

BANK Tool: List every way **Triangles by size** (L4-FAN-V)

The top point is joined to 5 points on the base line. How many triangles?

Answer. 10 (4 + 3 + 2 + 1).

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Triangles by size).

Ask. How do you know you have not missed one or counted one twice?

Listen for. An ordered list BC, BD, BE, CD, CE, DE, or a count by size 1, 2, 3. Listen for a rule such as "choose 2 base points".

Lesson 9: Data Detectives

SLIDE Tool: Draw it **Spinners with unequal parts** (L4-SPIN)

A spinner is split into 3 parts. Yellow takes 1/2 of the circle. Green takes 1/3 of the circle. Blue is the rest. (a) What fraction of the circle is blue? (b) Which colour is the spinner most likely to stop on? (c) Mei spins 60 times. About how many times should she expect yellow?

Answer. (a) 1/6. (b) Yellow. (c) About 30 times (green about 20, blue about 10).

Say. The picture is a fair way to see the parts of one whole circle. Think of the whole as 6 equal slices.

Ask. How many slices is yellow? Green? How many are left for blue?

Listen for. Common denominator 6: 3/6, 2/6, 1/6. Listen for "more of the circle means more likely". For (c) accept "about", never "exactly".

Lesson 10: Puzzle Day

SLIDE Tool: List every way **Timetables, trying both orders** (L4-TIME)

Kumar leaves home at 2:40 pm. He must go to the post office and the library, in either order, and then walk home. He stays 10 minutes at each place. The post office closes at 3:30 pm. The library opens at 3:10 pm. Walking times: home to post office 20 minutes, home to library 30 minutes, post office to library 15 minutes (each way). (a) Which place should he visit first? (b) At what time does he get home?

Answer. (a) The post office first. (b) 4:05 pm.

Say. There are only two orders. Try the first order, write the clock times down the page, then try the other order.

Ask. In the library-first order, what time does he reach the post office? Does it work?

Listen for. Library first: arrives 3:10, leaves 3:20, reaches the post office at 3:35, too late. Post office first: 3:00, leaves 3:10, library at 3:25, leaves 3:35, home at 4:05.

BANK Tool: List every way **Timetables, trying both orders** (L4-TIME-V)

Change the post office closing time to 3:40 pm. Which orders work now, and which gets Kumar home sooner?

Answer. Both orders work. Post office first gets home at 4:05 pm; library first: library 3:10 to 3:20, post office 3:35 to 3:45, home 4:05 pm. They tie at 4:05 pm.

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Timetables, trying both orders).

Ask. In the library-first order, what time does he reach the post office? Does it work?

Listen for. Library first: arrives 3:10, leaves 3:20, reaches the post office at 3:35, too late. Post office first: 3:00, leaves 3:10, library at 3:25, leaves 3:35, home at 4:05.

SLIDE Tool: List every way **Smallest and largest totals** (L4-DICE)

Siti rolls 4 ordinary dice (faces 1 to 6) and adds the 4 scores. (a) Find the least total possible. (b) Find the greatest total possible. (c) Which of these totals can never happen: 3, 4, 13, 24, 25?

Answer. (a) 4. (b) 24. (c) 3 and 25 are impossible.

Say. Do not roll anything. Ask what the lowest face and the highest face on one die are, then think about 4 dice.

Ask. Can she get every total from 4 to 24? How can you tell?

Listen for. Smallest 1 + 1 + 1 + 1, largest 6 + 6 + 6 + 6. Listen for "change one die by 1 at a time", which gives every total in between.

BANK Tool: List every way **Smallest and largest totals** (L4-DICE-V)

Siti rolls 5 dice and adds the scores. Find the least and the greatest total possible.

Answer. 5 and 30.

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Smallest and largest totals).

Ask. Can she get every total from 4 to 24? How can you tell?

Listen for. Smallest 1 + 1 + 1 + 1, largest 6 + 6 + 6 + 6. Listen for "change one die by 1 at a time", which gives every total in between.

SLIDE Tool: List every way **Counting in two stages** (L4-STAGES)

Aisha owns 4 tops, 3 skirts and 2 hats. An outfit is 1 top, 1 skirt and 1 hat. (a) Find the number of different outfits. (b) The red top never goes with the green skirt. Find the number of outfits that are left.

Answer. (a) 24 outfits. (b) 22 outfits.

Say. Draw 4 boxes for the tops. Under each one, show 3 skirts. How many top-and-skirt pairs is that?

Ask. How many outfits use the red top with the green skirt? What do you do with that number?

Listen for. 4 × 3 = 12 pairs, times 2 hats = 24. Listen for "multiply the choices". For (b) the red-green pair has 2 outfits (one for each hat); 24 − 2 = 22.

BANK Tool: List every way **Counting in two stages** (L4-STAGES-V)

A bus route has 3 ways from the depot to the market and 4 ways from the market to the harbour. Find the number of routes from the depot to the harbour through the market.

Answer. 12 routes.

Say. Same skill, new numbers. Use the same tool and the same steps as the slide item (Counting in two stages).

Ask. How many outfits use the red top with the green skirt? What do you do with that number?

Listen for. 4 × 3 = 12 pairs, times 2 hats = 24. Listen for "multiply the choices". For (b) the red-green pair has 2 outfits (one for each hat); 24 − 2 = 22.